ZHANHONG HE

Audio and Music Machine Learning Researcher

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Profile

Early-career researcher in audio and symbolic music modelling, with industrial research experience at Huawei and Dolby. Research spans expressive dynamics, sequence modelling, generative models and human perceptual evaluation. Hands-on experience with Python/PyTorch, Transformers, VAEs and preference-based post-training.

Research Experience

Research Intern | Huawei Hongkong Research Centre

Appointment began June 2026

- Worked on preference-based post-training for music-generation models using DPO and GRPO.
- Designed rollout workflows and music-oriented rewards informed by signal processing and music information retrieval.
- Collected and analysed human-preference data for music generation.

Research Intern | Dolby Laboratories, Sydney

May-September 2024

- Developed music-generation models using U-Net, VAE and masked Transformer architectures.
- Designed and conducted human listening tests with professional listeners.
- Contributed to research on inferring expressive MIDI velocity, including a U-Net approach to filling note-level velocity.

Doctoral Research | University of Western Australia

February 2022-present

- Studied expressive musical dynamics from audio and symbolic information, including MIDI velocity, dynamic markings and musical structure.
- Developed a multitask CRNN for dynamic markings, beats, downbeats and structural changes; reimplemented a Bark-scale specific-loudness feature extractor in PyTorch.
- Built a score-informed Transformer correction module to refine MIDI velocity estimates using score onset and frame information.
- Investigated cross-instrument velocity estimation using differentiable SoundFont proxies and audio-loudness supervision.

Visiting Student Researcher, part time | Great Bay University

Research visit began March 2026

- Explored pretrained music representations and agentic workflows for music-information-retrieval research.

Selected Publication And Projects

He, Z., Meng, H., Huang, D. and Togneri, R. (2026). Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-Task Multi-Scale Network. ICASSP 2026. DOI: 10.1109/ICASSP55912.2026.11460839.

Score-informed Transformer for Refining MIDI Velocity in Automatic Music Transcription

Research project on score-conditioned sequence refinement and expressive musical dynamics.

Filling the MIDI Velocity using U-Net Image Colorizer

Research project adapting image-colourisation methods to note-level velocity inference, with professional listening evaluation.

Education

University of Western Australia

PhD studies in Electrical and Electronic Engineering | February 2022-present

Defence completed; degree certificate expected November 2026.

University of Western Australia

BSc (Honours) in Engineering Science | October 2018-December 2021

First Class Honours | GPA: 7.0/7.0 | WAM: 87.7/100

Honours thesis: Improving Spectrogram-based Music Source Separation using Phase Information.

Technical Skills

Python; PyTorch; Transformer, CRNN, U-Net and VAE research; music-generation post-training; audio and symbolic representations; perceptual evaluation.

Teaching And Collaboration

Tutor and Lab Facilitator | University of Western Australia | February 2023-December 2025

Taught signal-processing and engineering-practice classes; co-supervised computer-science honours projects.